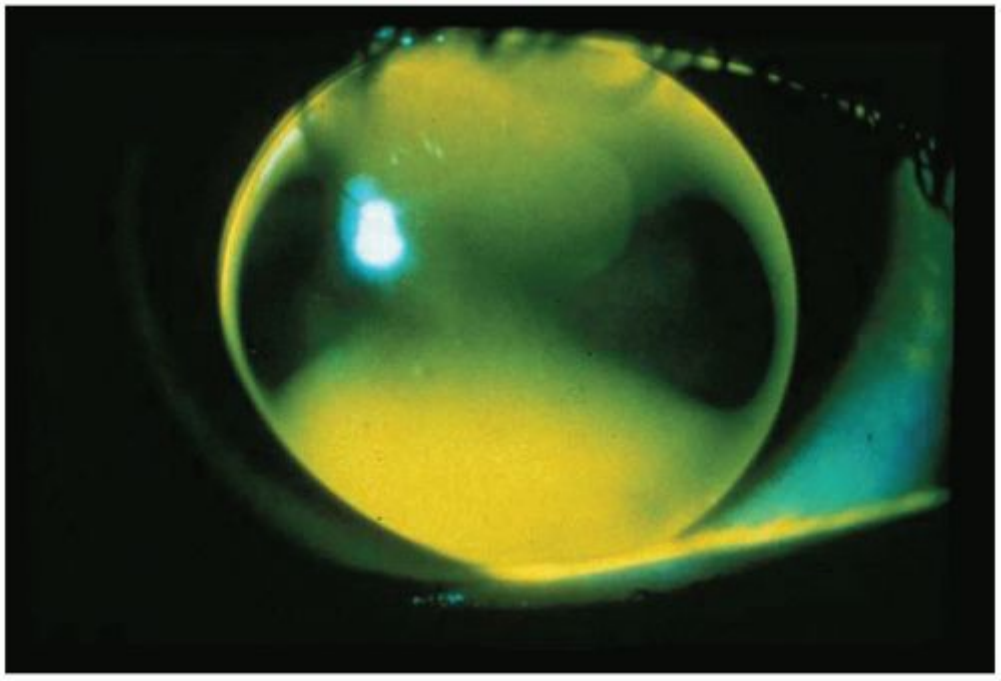
Section 1: Vinod's

1) What are the fitting characteristics, and what do we expect to see with: Rigid CL, K readings 7.80@180, 7.60@90, fitted with a BC of 7.60mm (Steep lens)(WTR)



- Slight apical clearance with superior and inferior clearance
- Dumbbell (3,9) appearance of touch

a) Rigid CL, K readings 7.60@180, 7.70@90, fitted with a Bc of 7.60mm (ATR astig)

- Slight apical clearance
- Slight 3 and 9 clearance
- Vertical dumbbell touch

b) Soft Cl, K readings 7.60@180, 7.70@90, fitted with BC of 7.60mm (Very tight lens)

- Very little movement on upgaze/blinking/lateral gaze
- Blanching of perilimbal blood vessels and conjunctival drag
- Vision/mires clears after blink (seen on keratometry)

2) . Questions about lens calculation like in the assignment. What is the BC, ICR, PCW and BVP, what is the lens design. There are 3 lenses. Then choose one of the 3 lenses, what would you do to make the fit looser. Give a rough estimation of how much to change the parameters by to give a small but clinically significant effect.

- <u>Base curve</u>
  - ➢ Radius of curvature of the central back optic zone of the lens
    - Rigid lenses ≈ central corneal curvature
    - Soft lenses – flatter than central corneal curvature (needs to fit the periphery of the eye)

- <u>Back vertex power</u>
  - ➢ Power (in dioptres) of the optic zone(s) of the lens measured in air – overall refractive power of the lens
  - ➢ Primarily determined by, and dependent on, the radius of curvature of the front surface of the lens

- Optic zone diameter (OZD)
  - ➢ Larger than maximum pupil diameter → measure under low illumination
  - ➢ Limited by other lens parameters (e.g. LD)

- Peripheral curve
  - ➢ Specified by:
    - Peripheral curve width (PCW)
    - Peripheral curve radius of curvature (PCR)
  - ➢ $LD = OZD + (2 \times PCW)$
  - ➢ Peripheral curve radius is much flatter than central base curve
    i.e. $PCR >> BC$ (mm)

- Intermediate (secondary) curve(s)
  - ➢ Specified by:
    - Intermediate curve width (ICW)
    - Intermediate radius of curvature (ICR)
  - ➢ $LD = OZD + (2 \times ICW) + (2 \times PCW)$
  - ➢ Intermediate curve radius flatter than base curse, steeper than peripheral curve
    i.e. $PCR > ICR > BC$ (mm)
  - ➢ Intermediate curve diameter is the diameter of the intermediate zone (outside edge)
    $ICD = OZD + (2 \times ICW)$

TIGHTEN		LOOSEN
Increase	**LD**	Decrease
Increase	**BOZD**	Decrease
Decrease	**BOZR**	Increase
Decrease	**I/PCR**	Increase
Decrease	**Thickness**	Increase

Two ways to increase lens diameter:

1. Retain same OZD, increase width of PC system
   - ➢ This will maintain same central fit, increase edge width
   - ➢ Minimal effect on lens stability
2. Increase OZD, retain same PC system
   - ➢ Will affect both central and edge fit of the lens
   - ➢ Central curve will be over peripheral part which is flatter → will look like steeper fit

Changing optic zone diameter

- To retain the same fluorescein pattern:
  - ➢ For every 0.5mm change in BOZD need to change BOZR by 0.05mm
    $0.5mm \uparrow BOZD \rightarrow 0.05mm \uparrow BOZR\ (flatter)$
    $0.5mm \downarrow BOZD \rightarrow 0.05mm \downarrow BOZR\ (steeper)$

3) How do the tear film fluid forces interact with the rigid CL? What happens to these forces in excessive tearing?

- **Tear film meniscus**
  - ➢ Location:
    - At eye lid margins → meniscus at junction of corneal and eyelid upper and lower margins (lacrimal river, tear prism)
    - At rigid lens edge → meniscus formed all the way around the CL edge
  - ➢ Nature:
    - Concave = “thirsty” → draws tears from surrounding areas
    - Creates a negative (“suction”) pressure under lens, both menisci pulling fluid off front surface of eye, front surface of CL and from underneath the lens

- The pressure pulling from underneath the lens is an important reason for why the lens stays on the eye → pressure determines how strongly the lens is held against the cornea

- **Friction** – interaction between lens and cornea → dependent on viscosity and thickness of post-lens tear film
  - ➢ Thinner layer → less movement
  - ➢ More viscous layer → less movement
  - ➢ Very thin layer → lens adherence

- **Negative pressure**
  - ➢ Larger R
    e.g. watering eye – lens lifted away from cornea
    - Less negative pressure
    - Less adherent lens
    - More mobile lens
  - ➢ Smaller R
    e.g. dry eye – lens edge close to cornea
    - Greater negative pressure
    - More adherent lens
    - Less mobile lens

4) Advantages and Disadvantages of apical touch, apical clearance fittings, and alignment fit.

Achieving alignment fit:

1) **Lens design**
   - Multicurve designs
2) **Lens diameter**
   - Based on corneal diameter (~HVID), palpebral aperture size, eyelid position and tightness
   - 8.8mm to 9.6mm
3) **Optic Zone diameter**
   - Based on MAX pupil size
   - <5mm =7.6mm OZD
   - 5-7mm = 7.8mm OZD
   - >7mm = 8.0mm
4) **Base curve**
   - Depends on level of toricity

Corneal toricity	**Base cure**
Spherical	On K
0.25 to 1.00	On K or 0.25D > K
1.00 to 2.00	0.50D > K
>2.00	ΔK/3

Fitting philosophy	Advantages	Disadvantages
Alignment fit "On K"	• Even weight distribution • Allows good tear circulation behind the lens • Minimises corneal distortion, spectacle blur • Has 'stood the test of time'	• Occasional poor centration • Lenses can ride high - due to interaction with upper lid • Moderately large lens movement - limbal indentation • Fit can flatten- due to steepening of cornea with oedema.

Apical clearance "Steeper than K" (for aussies)	• Better centration - good for wide palpebral aperture and hyperopia	• Poor tear exchange • Changes to corneal shape
Apical touch "Flatter than K"	• Aids with lid attachment • Good tear exchange	• Low riding if not supported by the upper lid • Excessive lens movement

5) What are desirable features of contact lenses? Compare and contrast your answer while discussing the differences between PMMA and HEMA, (Added in RGP as well)

Desirable features of contact lenses:

- Optical clarity
- Biocompatibility
- Oxygen permeability
- Durability
- Dimensional stability
- Hardness
- Wettability
- Resistance to deposition / easy to clean
- Ease of fabrication
- Low cost

Type of contact lens	Desirable qualities	Disadvantages
PMMA -Poly methyl methacrylate 1940s	- Had most of the desirable qualities in the list above - Better tear exchange than SCLs	- No oxygen permeability. - Dk = almost 0
RGP - Rigid Gas Permeables 1970s	Most of the above qualities **Gen 1 - Silicone acrylates** - Siloxane added to increase Dk to max 30 - MMA for mechanical strength **Gen 2 - Fluorosilicone acrylates** - Fluorinated monomers increased Dk further to max 60, - Also increased deposition resistance - UV protection incorporated **Super / Hyper Dk materials (developed 2000s onwards**) Better tear exchange than SCLs	**Gen 1 - Silicone acrylates** - Siloxane decreased wettability which increased amount of deposition **Gen 2 - Fluorosilicone acrylates** - Lipid deposition still a problem
HEMA - 2 hydroxymethyl methacrylate (Hydrogel)	- Comfort - Minimal effect on corneal curvature Silicone Hydrogels - High Dk	- Vision is not as good - Easy to damage - Limited lifespan - Easily contaminated - Requires more rigorous care regimen - Attracts deposits and are difficult to remove - Related to ionicity **Hydrogels**: - Dk is directly related to water content

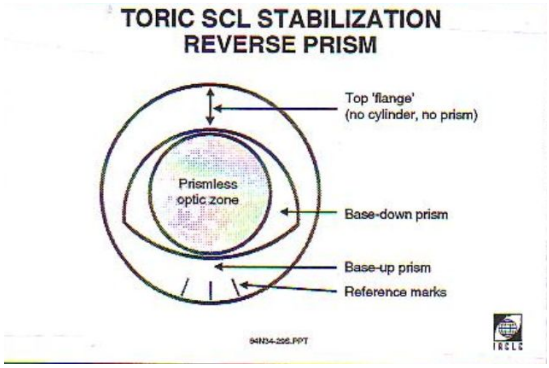
6) What are the initial anterior eye measurements you would take before fitting a patient. What is the significance of taking these measurements? What do they tell you? How does it affect your fitting?

- **Central corneal K**
  - Tells you whether there is any corneal toricity i.e. whether to use spherical or toric BC
  - Selection of base curve to:
    - Allow adequate tear exchange (BOZR and peripheral curve selection)
    - Avoid excess pressure/lens bearing
    - Avoid excess lens movement
- **Overall corneal shape**
  - Regular/irregular (keratoconus)
  - Helps choose peripheral curve design to maintain corneal alignment e.g. how many peripheral curves necessary to match corneal topography and their radii of curvature
  - Specialty lens e.g. aspheric or Conoid lens design
- **HVID/ Corneal diameter**
  - Required diameter of Hard/Soft CL
  - RGPs should be less than corneal diameter and SCLs should be more than corneal diameter
  - Hard = HVID - 2 RULE: 10mm HVID = 8.8 LD, 10.5 = 9, 11 = 9.2 etc
  - Soft = HVID + 2
- **Pupil size**
  - Make sure vision is not affected by pupil size, OZD = pupil in a dark room + 1mm
- **Palprebral apeture**
- **Lid tightness**
- **Lid position**
  - Can suggest lid attachment/ position of lens

7) What methods are used to keep toric lenses on axis of the eye? Described each method.
8) What are lens stabilisation techniques in soft and rigid lenses?

Hard Cls do not need lens stabilisation if fitted on K. If fitted flat, can use prism ballast

Prism ballast	- 0.75 - 2D BD (BOTTOM IS THICKER) - Created by offset lathing of two surfaces - Upper lid forces during blink maintain lens orientation - Watermelon seed Disadv - Thick lower lens edge > reduced dk/t - Discomfort from lower lens edge
Peri-Ballast	- No prism is optic zone - Negative lenticulation of peripheral zone - “Slab-off” of lenticulation in upper portion of lens give thin upper edge - Watermelon seed principle maintains lens orientation - Thinner lens overall cf prism ballast Disadv - Thick lower lens edge > reduced dk/t - Discomfort from lower lens edge
Dynamic Stabilisation MOST COMMON(thin zone, double-slab off, reverse prism)	- Thin zones at top and bottom of lens, thicker laterally - Thinner lens = more comfortable - Lid forces during blink maintain lens orientation

Lens Truncation LAST RESORT	- 1-1.5mm truncation at bottom (cutting the lens) - Align with lower lid margin for lens margin interaction - Normal or high lower lid preferred - Lid relatively tight - Comfort can be compromised due to lens edge
Reverse prism 	- Looks like it's hard to make - Prismless optic zone - Just below at 5 and 7 o clock is BD prism - Below the BD prism is base up prism - Should work similar to dynamic stabilisation and prism balast - "Watermelon principal" *err only pic provided by Vinod* *just guessed the properties*

9) Calculations for ordering a toric lens (rigid and soft)

When fitting the Spherical RGP

- Patient selection
  - ➢ <2.00DC corneal, low amounts of residual cyl
- BOZR

CORNEAL ASTIGMATISM	BOZR (D)
0.00 – 1.00D	On Flat K
1.12 – 2.00D	¼ ΔK + Flat K
2.12 – 2.87D	⅓ ΔK + Flat K
>3.00D	BST

- BVP
  - ➢ Adjusted for the **tear lens** created on **flat K** (SAM-FAP rule)
  - ➢ We are **steepening** the **flat K** → add minus of difference between D of flat K and BOZR fitted for BVP

- Example:

  Spectacle Rx (12 mm vertex)
  -4.50/-2.00x180
  Ks
  42.50 @ 180, 44.25 @ 090. Fit with spherical RGP
  Ocular Rx
  -4.25-1.75x180
  BOZR
  (Fit guide says fit ¼ ΔK + Flat K)
  -1.75D /4 + 42.50 = 42.9375 = 43.00 D/7.85 mm
  BVP
  -4.25 + (-0.50D Tear lens) = -4.75D

When fitting the Front Surface Toric RGP [not usually fitted, soft lens much easier]

- Patient selection
  - ➢ Low corneal astigmatism, significant residual astigmatism
- BOZR

CORNEAL ASTIGMATISM	BOZR (D)
0.00 – 1.00D	On Flat K
1.12 – 2.00D	¼ ΔK + Flat K
2.12 – 2.87D	⅓ ΔK + Flat K
>3.00D	BST

- BVP
  - ➢ As cyl is on **front surface**, need to specify in **+ve** cyl form
- Lens specific modifications
  - ➢ Needs a **method of stabilization** → using prism ballasting in the form of **base down prism** (usually between 1-1.5Δ)
    - Request lab to dot base
  - ➢ On average, prism tends to **rotate 10-15 nasally** (CCW OD, CW OS) → **compensate** with initial lens ordering using rotational rule (LARS)
    - If lens rotates **left** → **add**
    - If lens rotates **right** → **subtract**
  - ➢ Effect of prism
    - Prism will cause an increase in CT of lens

      $Prism\ CT \times 100 = prismatic\ power \times overall\ diameter\ (OAD)$

- Example

  Rx

  OS -2.50/-1.25x090

  Ks

  43.00@090, 43.25@180

  Residual Astigmatism

  -1.25 – (-0.25) = -1.00Dx090

  Diagnostic Lens

  BOZR 43.25D, BVP -3.00D

  Over refraction

  +0.25-1.00x090

  BVP

  Diagnostic Lens + O/R → Target: -2.75/-1.00x090

  Since this is FST, need +ve cylinder form = -3.75/+1.00x180

  Assume 15 CW Rotation = -3.75/+1.00x015 – Also need prism 1Δ

  Thickness

  Prism of 1Δ in a 9 mm OAD lens = 0.09 mm extra thickness in centre

  If standard lens CT is 0.15 mm, CT with 1Δ prism is 0.24mm

PARAMETER	
BOZR	43.25D (7.80)
BVP	-3.75/+1.00x15
OAD	9.0
OZD	7.80
SCR/W (=BCR + 1mm)	8.8/0.3
PCR/W (=SCR + 2mm)	10.8/0.3
CT (0.0mm for prism)	0.25
Prism	1 Δ, dot base

- Troubleshooting
  - ➢ Use the LARS rotational rule to compensate for rotation → reorder lens

➢ More prism and truncation may be needed to stabilize lens

When fitting the Back Surface Toric RGP

- Patient selection
  - ➢ Large amounts of corneal astigmatism
  - ➢ Refractive astigmatism > corneal astigmatism in **same axis**
- BOZR
  - ➢ Alignment fit
    - The two BOZR of the lens are chosen to match the two Ks of the cornea
    - **Fit** and **stability** against rotation maximized
    - Induced astigmatism **maximized**
    - **Prevents tear exchange**
    - The fluorescein pattern with mimic the pattern seen with an **spherical alignment fit** on a **spherical cornea**
  - ➢ Low toric stimulation (more commonly used)
    - The BOZR are chosen **slightly flat for both the flat and steep meridians** of the cornea
    - This allows for better **tear exchange** underneath the lens, while still providing adequate rotational stability so is more commonly used
    - The fluorescein pattern will **mimic** the look of a **spherical RGP** on a **low toric cornea**

CORNEAL TORICITY	FLAT MERIDIAN (Mandell-Moore)	STEEP MERIDIAN (Mandell-Moore)
2.00D	On K	0.50 Flatter
2.50D	0.25 Flatter	0.50 Flatter
3.00D	0.25 Flatter	0.75 Flatter
3.50D	0.25 Flatter	0.75 Flatter
4.00D	0.25 Flatter	1.00 Flatter
5.00D	0.25 Flatter	1.25 Flatter

The fit factors are suggestions – they are a starting point in the selection of BOZR in the design of a BST lens
You may/will have to modify the lens parameters when you actually fit the lens to the patient

- Peripheral curves
  - ➢ Secondary Curve Radii (SCR)
    - Generally 1.0 mm flatter than the BOZRs
  - ➢ Peripheral Curve Radii (PCR)
    - Generally 3.0 mm flatter than the BOZRs/2.0 mm flatter than SCRs
  - ➢ By specifying two SCRs and PCRs, you will have a
    - Results in a circular optic zone, and is preferred by many clinicians as it is more stable against rotation
- BVP
  - ➢ Modified flat meridian

- Example

Ocular Rx
Pl/-3.75x090
Ks
42.00 @ 090, 44.75 @ 180
Residual Astigmatism
-3.75D – 2.75D = -1.00D
BOZRs
Mandell Moore designed toric base curves – 41.75 @ 090, 44.00 @ 180 – ΔKCL = -2.25 D
Induced astigmatism: 0.5 x -2.25 D = **-1.12D** (RA = 1.00D)

BVP

Flat meridian: Fit 0.25D Flat BVP = pl + (+0.25D) = +0.25D

PARAMETER	
BOZR	41.75/44.00 (8.08/7.67)
BVP	+0.25
OAD	9.2
OZD	7.80
SCR/W (=BCR + 1mm)	9.08/8.67:0.35
PCR/W (=SCR + 2mm)	11.08/10.67:0.35
CT (0.0mm for prism)	0.15
Prism	None

When fitting the Bitoric RGP [most common form]

- Patient selection
  - ➢ Large amount of corneal and refractive astigmatism
  - ➢ Induced astigmatism does not adequately/ appropriately correct residual astigmatism
- BOZR
  - ➢ Similar to BST

CORNEAL TORICITY	FLAT MERIDIAN (Mandell-Moore)	STEEP MERIDIAN (Mandell-Moore)
2.00D	On K	0.50 Flatter
2.50D	0.25 Flatter	0.50 Flatter
3.00D	0.25 Flatter	0.75 Flatter
3.50D	0.25 Flatter	0.75 Flatter
4.00D	0.25 Flatter	1.00 Flatter
5.00D	0.25 Flatter	1.25 Flatter

- BVP
  - ➢ Difference between Bitorics and BSTs is with specifying BVP
    - In BST **only one** BVP is specified
    - In Bitorics the BVPs in **both meridians** is specified
    - BVP in each meridian is specified relative to ocular Rx and the fit factor (SAM-FAP)
  - ➢ The BVP specified should be what you want to be read in the lensometer – compensated for the tear lens due to fit
  - ➢ The lab will correct for the induced astigmatism to get you the BVP specified

- Example

Ocular Rx
-6.00-3.00x180
Ks
42.50 @ 180, 45.50 @ 090
Residual astigmatism
-3.00 – (-3.00) = 0 D (So bitoric lens will be a SPE lens)
BOZR
Mandell Moore Designed – 42.25D (0.25 flat) and 44.75 D (0.75 flat)
BVP
Flat: -6.00 D + 0.25D = -5.75 D
Steep: -9.00 D + 0.75D = -8.25 D

PARAMETER	
BOZR	42.25/44.75 (7.99/7.54)
BVP	-5.75D/-8.25D
OAD	9.2
OZD	7.8
SCR/W (=BCR + 1mm)	8.99/8.54:0.35

PCR/W (=SCR + 2mm)	10.99/10.54:0.35
CT (0.0mm for prism)	0.15
Prism	None

10) Define "refractive astigmatism", "internal astigmatism", "corneal astigmatism", and "residual astigmatism". How are they clinically measured? Also discuss how these terms are used to help us determine what lens form is best for the patient.

- **Refractive astigmatism:**
  - Total astigmatism of the eye, can be done alone via subjective or objective refraction
  - Refractive astigmatism = Internal astigmatism + Corneal astigmatism
- **Corneal astigmatism:**
  - Astigmatism induced due to the shape of the cornea i.e. different curvature along different meridians of the cornea
  - Measure with a keratometer, topographer, placido disc, etc
- **Internal astigmatism**
  - Astigmatism due to the shape of the lens, retina, tilt of the lens, misalignment of ocular elements

- **Residual astigmatism**
  - Remaining astigmatism once the ammetropia has been corrected.
  - Indicating how much of the astigmatism of the eye is non-corneal related.
  - Most often 0.50D or less, mainly ATR

Example
Ocular Rx -3.00-3.50x090
(vertexed to cornea)

K 41.00@090, 43.00@180
(Keratometer)

Refractive astigmatism = -3.50 x 090
Corneal astigmatism = -2.00 x 090

Residual astigmatism is -1.50x090

Usefulness of these terms:

- Residual astigmatism tells us theoretically, how much astigmatism would be left in the lens after fitting a spherical spherical BOZR RGP
- Depending on the amount of astigmatism that is corneal, different types of fitting would be appropriate
  - **Spherical RGPs**
    - Good for corneal toricity <2.00D i.e. low residual astigmatism
  - **Front surface toric**
    - Good for cases where there is low corneal astigmatism BUT high residual astigmatism
  - **Back surface toric**
    - Because of the presence of INDUCED ASTIGMATIC CORRECTION
    - Good for high toricity corneas i.e. corneal astigmatism >2.00D
      - BUT, when refractive astigmatism is 1.2-1.4x corneal astigmatism
    - I.e. Good for cases where the induced astigmatism happens to correct the residual astigmatism.
  - **Bitoric lenses**
    - At least 2.00D of corneal astigmatism
    - SPE (Spherical Power Effect) Bitoric = when residual astigmatism is low
    - CPE (Cylindrical Power Effect) Bitoric = when residual astigmatism is not negligible

RGP LENS TYPE	CORNEAL ASTIGMATISM	REFRACTIVE ASTIGMATISM	ADVANTAGES	DISADVANTAGES
Spherical	Low <2.00DC	Minimal	➢Up to 90% of corneal astigmatism corrected ➢Simplest/least complicated fit	➢Harder to get good fit as corneal astigmatism increases ➢Excessive edge clearance in steep meridian – uncomfortable ➢Limbal and conjunctival injection – 3 and 9 o'clock staining ➢Abnormal blink patterns or head postures ➢Not all corneal astigmatism is correct ("only" 90%) ➢Lens may flex/fold alone flatter meridian – inducing more astigmatism ➢Does not correct residual astigmatism ➢Potentially variable acuity
Front Surface Toric	Low	High	➢Good for correcting residual astigmatism after corneal corrected by spherical back surface	➢Relies on a good corneal fit with a spherical BOZR ➢Needs a method of rotational stabilization
Back Surface Toric	>2.00DC	20-40% higher than corneal	➢Better fitting lenses on astigmatic corneas ➢As front surface is spherical, can be modified in office if needed	➢Minimum corneal astigmatism required ➢ "Difficult" to fit – number of visits, time ➢Induced astigmatism
SPE Bitoric	>2.00DC	≈ corneal cyl	➢Good fit between toric cornea and cornea BOZRs ➢Can correct residual astigmatism in addition to induced astigmatism ➢SPE – lens free to rotate	➢"Difficult" to fit – time, cost ➢More expensive than BST ➢Optics not as "crisp" as BST ➢CPE – lens not free to rotate, needs a method of rotational stabilization
CPE Bitoric	>2.00DC	≠ corneal cyl		

11) Comment on the relationship between water content, permeability, and transmissibility in RGPs and hydrogel lenses.

RGP:

(Dk = permeability, Dk/t = transmissibility)

- Increased permeability = increased transmissibility
- The Max Dk/t for RGP lenses has not been found but we do not really care because of the ability to produce hyper Dk/t materials that already allow enough oxygen to permeate without resulting in hypoxia

Hydrogel lenses:

- Increased water content = increased Dk/t (increased transmissibility)
    - Larger polymer pore size = higher water content
    - Higher water content = larger pore size - absorb more substances, more easily
- For hydrogel lenses, there is a maximum Dk/t it can have which is 75-80% of the Dk/t of water.
    - Because water is the carrier of oxygen in this case.
    - Anymore than 80% and the lens will be too fragile

Note: The higher the oxygen permeability the higher the permeability of CO_2 out in the atmosphere as well, to less CO_2 accumulation in the stromal causing stromal acidosis and morphological changes to the endothelium.

12) When fitting a hyperope with optical considerations, why are hyperopes "happy" and myopes "miserable"?

Fitting Hyperopes

The Issues	*Solutions*
- Contact lenses for hyperopes are heavier than myopic lenses. As a result, it will cause a shift in the centre of gravity when positioned on the eye with the lens-cornea relationship, resulting a low riding lens that drops down. - Thinner edges give rise to a thinner upper edge resulting in the absence of the watermelon seed principle; hence lens will not be picked up - Will also give rise to a thinner lower edge and thus be provided no support from the lids.	- Reduce weight by changing the design, material and diameter - Increase lens edge thickness by negative lenticulation in order to promote lens-lid interactions - Apical clearance fit in order to produce a tighter lens that will result in a better centration.

The result of low riding lenses can result in a poor tear exchange, flare from optic zone and corneal distortions.

Hyperopes vs. Myopes (Happy hyperopes and Miserable myopes)
There are 2 aspects of wearing contact lenses where hyperopes will have an advantage over myopes:

1) **Convergence:**
    - For hyperopes wearing plus SPECTACLE lenses, when converging they have a BO prismatic effect induced meaning they perceive the object closer to them, thus requiring them to CONVERGE MORE WITH SPECTACLES
    - For myopes wearing minus SPECTACLE lenses, when converging they have a BI prismatic effect induced meaning they perceive the object to be further away from them, thus requiring them to CONVERGE LESS WITH SPECTACLES
    - **Therefore, by changing to CLs, Hyperopes with converge less but myopes will have to converge more (no more prism effect like in spectacles)**

2) **Accommodation**
    - Ocular accommodation (i.e. at the eye e.g. emmetropes and CL wearers) = +2.47D
    - +10 hyperopes with spectacle correction needs to accommodate +3.29D for same distance
    - -10 myope with spectacle correction needs to accommodate +1.83D for same distance
    - This is due to the changes in vergence since spectacle refraction takes place at a certain "vertex distance" away from the eye
    - **Therefore with spectacles, hyperopes accommodate more whereas myopes accommodate less**
    - Switching to CL would have the opposite effect

13) What are the advantages and disadvantages of different edge profiles?

Lens edge: The extreme edge of the peripheral curve

- Junction between the front and back surface where the edge LIFT and SHAPE can interact with the eyelid
- Can affect:
    - Comfort
    - Lens position
    - Lens movement

a) Edge shape:

- Eyelid margins are as sensitive as the cornea so interaction with the lens edge is a major factor for the comfort or a contact lens
- Rigid lens edge shape determines comfort
    - The more **rounded** the side of the lens facing the eyelid, the more comfortable the lens
    - The more **blunted** the more uncomfortable

b) Edge lift (clearance)

- Axial edge lift - distance from the edge of the contact lens to the corneal surface measured parallel to the optic axis
- **Aim for 60 to 90µm**
    - **Too high:**
        - Lid sensation → discomfort
        - Poor centration, excessive movement,
        - 3 and 9 o'clock staining due to meniscus-induced tear thinning and bridging effect
    - **Too low:**
        - Less lid interaction - may improve comfort
        - When lens decentres, may exert pressure
        - Can lead to epithelial/limbal trau
        - Can reduce or increase 3 or 9 o’clock staining
- Find best compromise for patient

c) Edge width

- Width of the peripheral band of the edge lift - usually relates to the width of the peripheral curve
- **Aim for 0.25 to 0.30mm**
    - **Too wide:**
        - Debris accumulation
        - Foreign body sensation
        - Increased lens movement, poor centration
    - **Too narrow**
        - Insufficient tear exchange
        - Oedema, accumulation of debris
        - Edge and indent limbus on blink
        - Discomfort
- Find compromise for patient

Section 2: Swarbrick's

1) Compare the physical/mechanical, metabolic and inflammatory effects that occur with rigid lenses compared to soft lenses.

		Rigid lenses	SCL
Physical	Corneal shape changes	• Cornea tends to mould to back surface • PMMA lenses → cause central edema → central steepening • High Dk/t RGP: minimal corneal shape change • Low Dk/t RGP: more marked corneal shape change cf high Dk/t • Lens decentration may cause that corneal area to change shape • May result in corneal warpage (chronic wear)	• Changes not as common/minimal
	Corneal sensitivity	• **Sensitivity decreased = CTT (Corneal Touch Threshold) increased** → Reduced corneal damage awareness • **CTT increase is dependent on Dk/t** • RGP lenses have less effect on CTT than PMMA	
	Eyelid sensitivity	• Less comfortable than SCL • Comfort dependent on lens edge shape and its interaction with eyelid margin • Sensory adaptation to rigid lenses occurs 7-14 days • However, 25-30% of people can't adapt	• More comfortable than rigid lenses as it tucks underneath eyelids
	Tear film	Pre-Lens Tear Film: • Less stable pre-lens tear film cf SCL (BUT: 4-6 seconds; thickness: 2.25µm) • Lipid layer rare Post-Lens Tear Film: • Greater provision of tear bourne oxygen cf SCL • Thicker than SCL 3 and 9 o'clock staining: only seen in rigid lens wearers • Due to dry of peripheral cornea outside rigid lens edge • Excessive edge lift - 'thirsty' meniscus	Pre-Lens Tear Film • BUT 8-12 seconds; thickness: 3-4µm • Occasional lipid layer Post-lens Tear Film: • Less tear borne oxygen cf rigid

		● Bridging effect at lens edge	
Metabolic	<u>O2 supply</u>	● Tear pump allows circulation of oxygenated tears behind lens ● Dependent on Dk material, and thickness of lens ○ O2 transmission promoted by polymers ○ Silicone acrylate polymer → O2 is transported through rotation of molecules ○ Fluorine based polymers PMMA lenses: - Dependent on tear film alone for O2 RGP lenses: - Dependent on tear film and diffusion through lens	● Minimal tear pump ● <u>Hydrogel lenses:</u> Dk is proportional to H2O content of polymer ● <u>Silicone hydrogel:</u> Dk increases as H2O of polymer is reduced Hydrogel and SiHy lenses: - Dependent on diffusion through lens only for O2
	<u>Corneal Edema</u>	● Edema confined to central 4-6mm of cornea <u>PMMA lenses:</u> ~ 6% central daytime edema <u>RGP:</u> <<6% daytime edema	● Edema induced across whole cornea ● <<6% daytime edema
	<u>Hypoxia</u>	Dependent on Dk/t - The lower the Dk/t, the more likely to induce hypoxic effects - This is as low Dk/t results in less oxygen supply to cornea → increases anaerobic metabolism → increases lactate production → becomes strong osmotic stimulus **FOR HYPOXIC EFFECT LOOK AT Q7 SWARBRICKS**	
Inflammatory effects	<u>Vascular Responses</u>	Rare	● More prevalent in hydrogel wear ● Increased limbal hyperemia ● Can cause corneal vascularisation - Superficial vascularisation most common complication - Deep stromal vascularisation is less common BUT could result in central scarring, reduced vision ● CLARE
	<u>Infiltrative Responses</u>	Rare	● More prevalent in SiH wear ● Can be due to solution preservatives, poor lens fit/condition, toxic environmental stimuli or bacterial endotoxins/exotoxins ● SLK ● CLPU
	<u>Papillary conjuncti</u>	● Hydrogel, SiH >> rigid lenses ● Rigid and SiH lenses tend to induce localised CLPC → suggests	

	vitis	mechanical etiology as these are stiffer materials ● Hydrogel more often associated with generalised CLPC

2) Diseases, causes, s&s, management

FOR THESE 3 QUESTIONS (i.e. DISEASES) CLICK ON THE LINK BELOW

https://docs.google.com/document/d/1854y_RLIF0b8IjGegdgLuv6ra2uflUHQ7JdGSxBN340/edit?usp=sharing

3) What are the risk factors associated with infection and what strategies could you use to minimise these?

Modifiable risk factors

Risk factors	**Strategies to minimise**
6-7 days/week of CL wear	
Extended wear	Switching to disposable dailies is an alternative to extended wear lenses
Internet purchase	Patient education - stress to patient that CLs purchased over the internet are more likely prone to infections
Poor case hygiene	Tell patient to bring CL case in at every aftercare visit → you can examine, and also replace with new CL case Give out new CL cases with solution purchase - however most of the time, CL cases are already included in the box Can also provide antimicrobial lenses cases → come with silver coating Can do pre-paid lenses - send out lenses when it is time to replace them Patient education on risk of microbial growth in lens case
3-5days/week CL wear	
Swimming in CL	Patient education: Explain to the patient the risks associated to swimming in CL → Acanthamoeba keratitis Suggest alternative solution: custom made goggles?
Overnight wear	Patient education: Stress to patient risks of bacterial growth

- Also advise patient to never rinse with tap water
- Rinse only with saline!!
    - Also warn patient about unpreserved saline → can very quickly become contaminated
- Poor compliance to lens hygiene routine
    - Switch to less complex if applicable? E.g. multipurpose solution

4) Instructions for new contact lens wearers including the lens cleaning procedure.

Instructions for putting them in:

- Make sure you wash your hands prior to handling of the lenses
    - Avoiding infections
- Make it a habit to insert the right lens first - minimises insertion into the wrong eye

- Check to see if the lenses are inside out.
- When putting contacts in, DON'T do it in the bathroom,
    - Do it on a flat table surface over a bowl, with a good mirror, good lighting, and a towel underneath
    - Don't worry if it drops as long as the lens is rinsed thoroughly with saline solution before it is inserted into the eye, that is fine. If dropped on the ground, be careful where you step.
- At the end of the day, dispose of them. DO NOT REUSE
- Spare lenses - for broken lens

Important Rules:

- DO NOT SLEEP IN THEM - Naps that are less than an hour is fine
- NEVER CLEAN WITH WATER
- NO SWIMMING - Unless you use goggles or dispose of them right after

Warning signs:

- Pain, redness, blurry vision, tearing, or discomfort during lens wear:
    - Use re-wetting eye drops to lubricate the eye and self lid massage
    - Otherwise, take out CLs, rinse with saline and put back in the eye
    - If symptoms persist, take out lenses and see the nearest optometrist
    - HOWEVER, if symptoms persist WITHOUT wearing of lenses, see the nearest optometrist IMMEDIATELY

After initial fitting - Aftercare appointment

- 2 weeks after wearing lenses, reason:
    - See how things are going
    - Review the way the lenses are being handled
    - Chance for patient to bring up questions or concerns
    - If all is good, next aftercare will be 12 months

EXTRA: FOR NON-DAILY CL WEARERS - Cleaning procedure

- Take out lenses are night, rub with cleaning solution for 10s on each side in a circular motion with index finger and all around the edges of the lens.
- Rinse the lens well before placing it into the lens case
- Remember to re-fill the lens case with new lens solution every night!
- Tightly seal the case.
- Next morning, repeat the cleaning process and insert your CLs
- Pour out all the solution from the lens case and use a tissue to cover your index finger and gently rub around the inside of the lens container - REMOVE SLIME BUILD UP
- Rinse the container with saline and leave it and the lids face down on a hand towel to dry throughout the day
- Replace the lens case every 3 months minimum
- NEVER keep the solutions past expiry day and discard 1 month after opening or as specified
- NEVER top out the solution in the container, always pour it out.

For the non-compliant person:

Refit to daily disposables

5) What are the adverse effects caused by non-compliance? How can you intervene to reduce these effects from occurring?

Adverse effects caused by non-compliance:

- Microbial keratitis
- Damage to lenses - change of lens parameters making it useless.

Contact lens non-compliance – common px mistakes

- Irregular use of daily cleaner
- Inadequate technique with daily cleaner
- Inappropriate rinsing technique
- Irregular changing of disinfectant solution
- Irregular cleaning of lens case
- Irregular hand washing

Solutions:

- Careful px selection – say NO to non-compliant px's
- Px education and in-office demonstration – ensure px understand the system (cleaning system, rub and rinse method)
- Use a simple lens care system – multi-purpose solution
- Improve communication – instructions should be clear and succinct, provide written instructions – laminate so it lasts forever!
- Review px cleaning technique of lens/ lens case and solution replacement schedules on EVERY aftercare visits
- Adjust lens replacement schedule AND aftercare visit frequency to suit the px (monthly, fortnightly or daily disposables)

6) What are acute and chronic hypoxic effects in the epithelium, stroma, endothelium, and how would you manage a patient with hypoxic effects?

General effects of hypoxia:

- The lack of oxygen leads to anaerobic metabolism which results in an increase in lactate
- CL also acts as a barrier for CO_2 to be released
- Buildup of lactate and CO_2 forms lactic and carbonic acid creating a strong osmotic gradient (hypertonic environment)
- This pressure gradient will draw water into the stroma from both the AC and the TF resulting in swelling

Epithelium:

a) **Acute effects:**
   - **Epithelial cellular changes:**
     - Accumulation of **fluid** and **lipid** like bodies in the cell
     - Distortion of the cell **shape**
   - **Compromised epithelial barrier integrity**
     - Tight junctions are loosened in hypoxia → loose junctions means more susceptible to infection
   - **Epithelial oedema**
     - Water gathering between epithelial cells ("Waterlogged") because of the increased lactate in the stroma creating a hypertonic environment resulting in water influx (from both AC and tear film)
     - Hazy and misty vision, halos around lights
   - Reduced corneal **sensitivity**
   - Reduce epithelial **mitotic rate**

b) **Chronic effects:**
   - Immediately after lens removal:
     - Epithelial thinning
     - Reduced oxygen uptake - epithelium did not need as much oxygen
   - Epithelial vacuoles
   - **Epithelial microcysts**

        - Indicates dead epithelial cells indicating hypoxic stress
        - **Intervene when >50 microcysts are seen OR microcysts are seen breaking through the epithelial surface** → creates access to stroma thus risking infection
    - **Epithelium integrity is reduced: predisposing epithelium to trauma and infection**
        - Reduced **adhesion** → reduced number of hemidesmosome
        - Increased **fragility**
        - Reduced corneal sensitivity → related to Dk/t
        - Older surface cells are retained for longer

Stroma:

- **Stromal thinning**
    - Slow leaching of proteoglycans from stroma due to acute/chronic oedema
- **Posterior stromal haze**
    - With high levels of hypoxia in metabolically susceptible individuals due to the accumulation of lactate
- **Edematous corneal formations (ECF)**
    - Cloud-like or stellate opacities in the mid-stroma
    - Formations of striaes and Descemet folds in oedema levels greater than 5% and 10% respectively

Endothelium

- **Endothelial blebs** - black spots with maximal response after 20-30min after lens insertion
- **Polymegethism** - cells of different sizes → quantified by coefficient of variation, grading scales
    - Cannot be reversed
- **Polymorphism** - cells of different shapes
- It has been found that **stromal acidosis** (due to a build up of carbonic and lactic acid) is a key factor in endothelial changes.

Management:

- Increase the amount of oxygen entering the cornea during contact lens wear by:
    - Changing lens material to an **increased Dk/t**
    - Thinner lens
- Wearing the contact lens less
- CCLRU study by Rotenburg showed epithelial hypoxic changes returned to normal in about 1-2 months
- Switching from EW to daily disposable SiH wear

7) Describe the range of available Dk/t for hydrogels, SiH and RGP, and what wear schedule they are appropriate for.

Thresholds to avoid oedema

Holden and Mertz 1984

1) To avoid daytime (open-eye) oedema,
   Dk/t = 24 (cm.mlO2)/(sec.ml.mmHg)

2) To limit overnight oedema to 8% (complete recovery during the day),
   Dk/t = 34 (cm.mlO2)/(sec.ml.mmHg)

3) To limit overnight oedema to 4% (equal to no-lens oedema)
   Dk/t = 87 (cm.mlO2)/(sec.ml.mmHg)

Sweeney et al 2003

4) To limit overnight oedema to 3% (equal to no-lens oedema)
   Dk/t = 125 (cm.mlO2)/(sec.ml.mmHg)

For reference, these are the values of the CLs I used from Assignment 2 (Leo)

- Boston IV: Dk = 19
- Boston XO2: Dk = 141
- Acuvue moist: Dk = 21.4
- Bioinfinifty: Dk = 128

Note that the above shows lens material Dk, not the Dk/t. To work out Dk/t of your lens, use:

t(critical) = Dk / (Critical Dk/t)

8) Advantages and disadvantages including patient characteristics for MPS, peroxide, and heat disinfection.

Disinfection:

A physical or chemical process which inactivates all vegetative organisms but not necessarily spores or cysts.

Method	***Advantages***	***Disadvantages***
Heating in saline for 10min at 80°C	• Very effective - except some bacterial spores, Acanthamoeba • No allergic reaction - unless preserved saline is used • Economical once heating unit is purchased	• Accelerates lens ageing • Denatures protein deposits • Can cause lens discolouration if preserved saline is used • Not suitable for HWC lenses (HWC lenses are fragile already) • Not portable for travelling
Hydrogen peroxide disinfection (oxidation)	• Very effective - time for exposure • No allergies	• Acute epithelial toxicity if not neutralised properly • Residual peroxide can lead to stinging, pH changes and cellular damage due to free radicals • Requires compliance and has a relatively long disinfection time
Chemical disinfection / Multi-purpose solutions	• Acceptable efficiency (but least effective) • Convenience • Low toxicity prior to insertion	• Adverse response/hypersensitivity due to the presence of preservatives • Long contact time of 4 hours - lenses may also absorb chemicals at this time • Compliance to rub lenses

9) Why are RGPs better than hydrogels?

RGP	***Hydrogels***
• RGP lenses allow for a much greater tear exchange with blinking as it enables a more efficient way of waste, debris and FB removal from under the lenses. This in turn has an indirect effect of reducing the chance of an inflammatory response. • RGP lenses also exhibit better tear exchange than hydrogel lenses (20% vs 1%), meaning that more oxygen is supplied to the cornea,	• Hydrogel lenses tend to absorb chemicals from cleaning/storage solutions and elements from the environment which may leach out during lens wear, causing an allergic response from the wearer. • Hydrogel lens wearers are at a higher risk of developing corneal infiltrates and infection. • Hydrogel lens wearers are more prone to dry eye issues.

hence reducing the risk of hypoxia related problems like conjunctival/limbal hyperaemia, corneal oedema, endothelial polymegethism and decreased epithelium integrity. ● RGP lenses are easier to clean and maintain, thus reducing deposit build-up problems which can cause allergic reactions and reduced vision.	

10) Disposable contact lenses are healthier for the eye than conventional hydrogel lenses. Discuss.

- Reduce exposure to allergens/inflammatory stimuli such as denatured protein deposits, chemicals from cleaning/storage solutions and other environmental contaminations.
- Disposable lenses are usually thinner than conventional lenses so they reduce the risk of adverse effects from physical interaction with the eyelids. E.g. CLPC, discomfort and inflammation. In some patients, it also reduces the amount of "plastic" in their eye.
- Disposable lenses have less opportunity to experience lens ageing; which is a phenomenon that causes the lens to fit tighter. Aged hydrogel lenses have reduced movement due to the tighter fit, causing tear exchange and accumulation of of debris and waste products in the post-lens tear film. They may also experience more frequent dehydration, which contributes to the lens degradation.

11) Patient complains about dry eye in the late of the day. What do we do?

Step 1: Dry Eye Assessment
Look for associated signs such as:
- Debris in tear film
- Short TBUT
- Irregular MG orifices
- Scaley lashes and crusty lid margins

Step 2: Treatment: (if due to MGD, or anterior bleph)
- Aim: To unblock the MG orifices and expressing the abnormal lipid from the glands
- Use of unpreserved ocular lubricants
- Lid hygiene:
    - **Warm compresses**
        - Soak a cloth in hot water, then squeeze out the water
        - Fold the towel and hold it on top of the eye for 1-2min
        - Repeat in the other eye
    - **Lid scrubs**
        - Fill a small basin with water and use 3-4 drops of Johnson's 'no tears' baby shampoo
        - Dip a cotton bud in the soapy water and gently roll the eyelid margin outwards
        - Rub the cotton bud softly along it
        - Get rid of any scaly or crusty parts on the eyelid margins and eyelashes
        - Repeat for the upper eyelid and in the other eye
    - **Gland expression:**
        - Performed in office for severe dry eye cases
        - Use a cotton bud press on the eyelid just below the lid margin and repeat for all orifices
        - Express as much of the contaminated secretion as possible until the MGD is resolved
    - Warm compresses and lid scrubs done by the patient at home
        - Morning and arvo

- Review in 2 weeks and continue treatment if needed

Step 3: Changing the lens

- Optimise the lens fit
- Change to a lower water content material or RGP → promotes tear pumping for lubrication
- Reduce the wearing time and wearing schedule
- More frequent after care visits

12) Differential diagnosis for mucin balls, microcysts and vacuoles

Mucin Balls	***Microcysts***	***Vacuoles***
Abolished on lens removal	Persist on lens removal	Persist on lens removal
Translucent	Reversed illumination	Unreversed illumination
NaFl pooling in depression	Occasional NaFl staining	No staining

13) Draw some common corneal staining see in optometric practice, and state the causes of the staining.

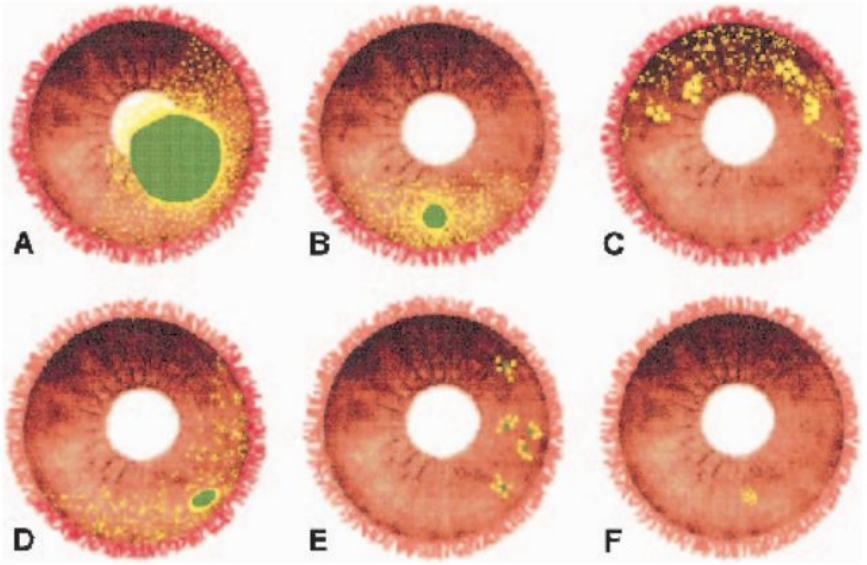


FIGURE 4. Schematic diagram outlining the six categories of infiltrative events seen with contact lens wear. **(A)** Microbial keratitis. **(B)** Contact lens–induced peripheral ulcer. **(C)** Contact lens–induced acute red eye. **(D)** Infiltrative keratitis. **(E)** Asymptomatic infiltrative keratitis. **(F)** Asymptomatic infiltration. Small yellow dots represent areas of diffuse infiltration, larger yellow dots represent focal infiltration, and green areas represent areas of corneal staining.

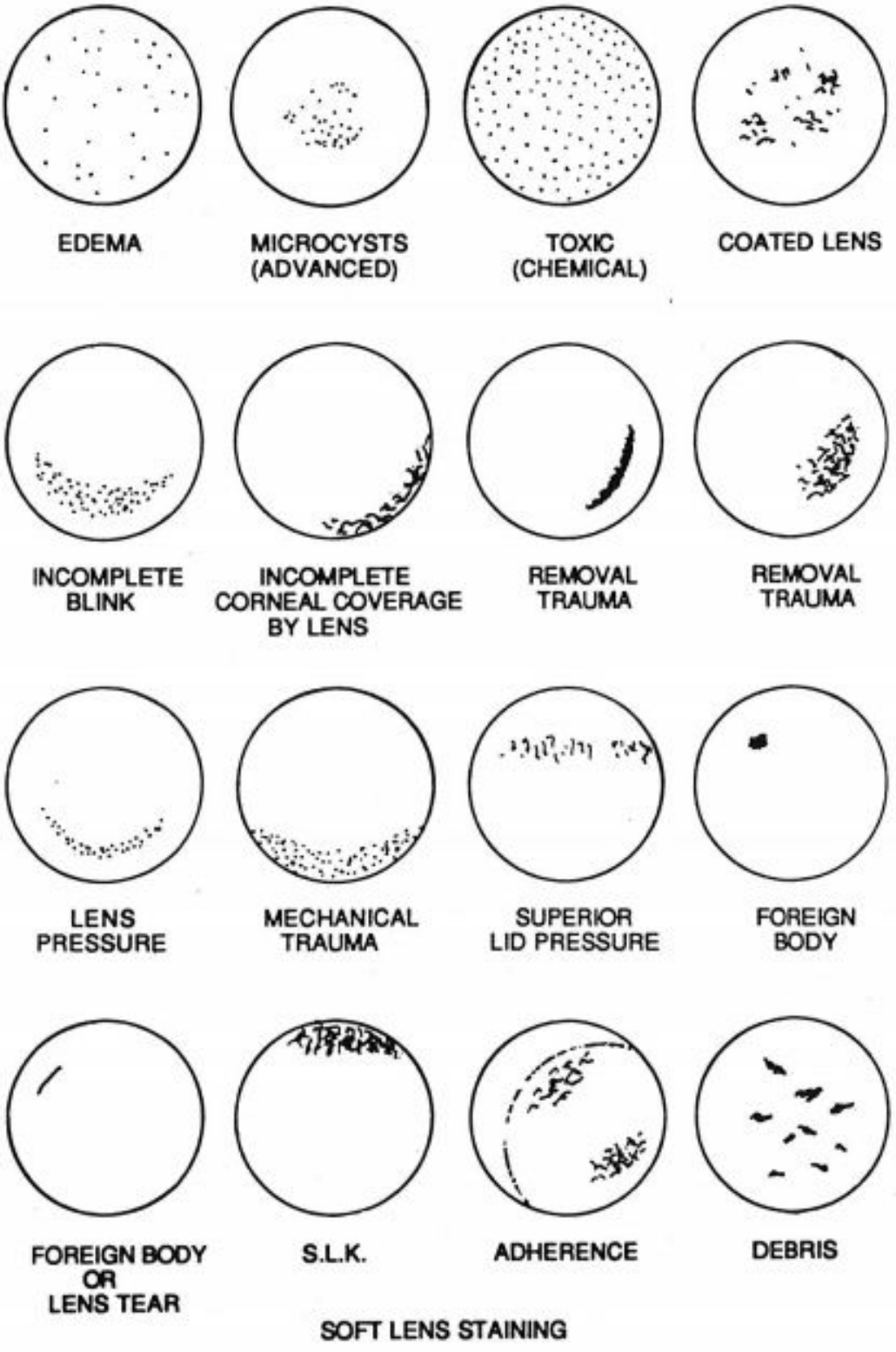


Figure 22.11c. Various types of corneal staining from gel contact lenses.

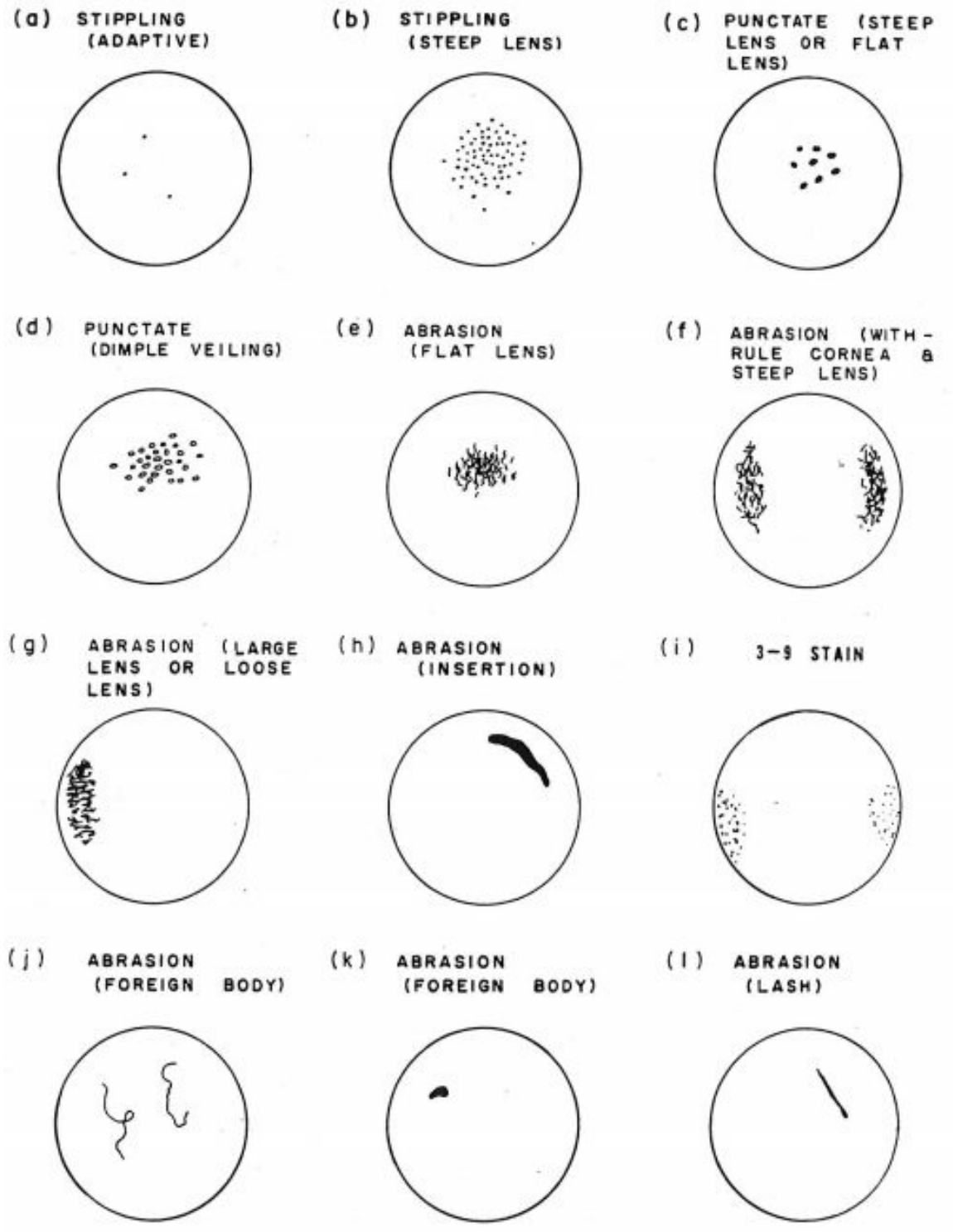


Figure 14.2. Illustration of various abrasions.